

Discrete analysis to ROBER PROBLEM

Not assuming QSSA

<1>

$$\begin{bmatrix} [A]_{t+1} \\ [B]_{t+1} \\ [C]_{t+1} \end{bmatrix} = \begin{bmatrix} 1 - k_1 \delta t & 0 & 0 & k_3 \delta t & 0 \\ k_1 \delta t & 1 & 0 & -k_2 \delta t & -k_3 \delta t \\ 0 & 0 & 1 & 0 & k_2 \delta t \end{bmatrix} \begin{bmatrix} [A]_t \\ [B]_t \\ [C]_t \\ [B]_t [C]_t \\ [B]_t^2 \end{bmatrix}$$

$$\begin{bmatrix} [s]_{t+1} \end{bmatrix} = \underline{\underline{U(s)}} \sum_{i=1}^n \underline{\underline{V_i(s)}}^T \begin{bmatrix} [rec]_t \end{bmatrix}$$

With QSSA assumed on B: $[B]_{t+1} = [B]_t$

<2>

$$\begin{bmatrix} [A]_{t+1} \\ [B]_{t+1} \\ [C]_{t+1} \end{bmatrix} = \begin{bmatrix} 1 - k_1 \delta t & k_3 [B]_t \delta t \\ 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} [A]_t \\ [C]_t \end{bmatrix} + \begin{bmatrix} 0 \\ [B]_t \\ k_2 [B]_t^2 \delta t \end{bmatrix}$$

Set Criteria for QSSA to be assumed:

1. given a species: $T_s = \frac{[s]}{[s]}$ $\frac{\delta t}{T_s} \leq \epsilon \Rightarrow \frac{[s]}{[s]} \cdot \delta t \leq \epsilon$

2. for species A : $i \rightarrow$ index for species, $j \rightarrow$ index for time

$$A_{j+1} = A_j + \frac{dA}{dt} \cdot \delta t$$

trying to find out how this 'ε' is going to affect other species.

<3>

$$\begin{aligned}
 &= A_j + \sum_{i=1}^m \frac{\partial A}{\partial s_i} \left(\frac{\partial s_i}{\partial t} \cdot \delta t \right) \rightarrow [s_i] \cdot \delta t \leq \epsilon [s] \\
 &\leq A_j + \sum_{i=1}^m \left(\frac{\partial A}{\partial s_i} \right) \epsilon [s] \rightarrow \text{untraceable analytically}
 \end{aligned}$$

try solving $\frac{dA}{ds_i}$ using analytical solution:

e.g. ROBER problem: $[A]_t = [A]_0 + \int_0^t -k_1[A] + k_3[B][C] \cdot dt$

$$\frac{\partial [A]}{\partial [B]} = 0 + \frac{\partial}{\partial [B]} \int_0^t -k_1[A] + k_3[B][C] \cdot dt$$



$$\frac{\partial [A]}{\partial [B]} = - \int_0^t \left\{ k_1 \cdot \frac{\partial [A]}{\partial [B]} + k_3 \frac{\partial}{\partial [B]} ([B][C]) \right\} dt$$

untraceable analytically

However, perturbation method can be applied to approximate $\frac{\partial A_i}{\partial A_j}$

Express <3> in a slightly modified way:

$$(A_{t+1} - A_t) = \sum_{j=1}^n \frac{\partial A_i}{\partial A_j} \cdot A_j \cdot \delta_t$$

↓ in a matrix way.

$$\begin{bmatrix} A_{t+1} - A_t \end{bmatrix} = \begin{bmatrix} 1 & \frac{\partial A_1}{\partial A_1} & \frac{\partial A_1}{\partial A_2} & \dots \\ \vdots & \vdots & \vdots & \ddots \\ \frac{\partial A_n}{\partial A_1} & \dots & \dots & 1 \end{bmatrix}_t \begin{bmatrix} \delta_t \cdot A_1 \\ \delta_t \cdot A_2 \\ \vdots \\ \delta_t \cdot A_n \end{bmatrix}$$

<4>

Reciprocal Matrix: $a_{ij} \cdot a_{ji} = 1$.

<5> proof otherwise ???

ROBER Example:

$$\frac{\partial A}{\partial B} \approx \frac{\Delta A}{\Delta B} = \frac{A(B+\Delta) - A(B)}{\Delta} = \frac{k_3 \Delta \delta_t \cdot C}{\Delta} = k_3 \cdot \delta_t \cdot C$$

$$\frac{\partial C}{\partial B} \approx \frac{\Delta C}{\Delta B} = \frac{K_2(B+\Delta) \delta_t - K_2 B \delta_t}{\Delta} \approx 2K_2 \delta_t B$$

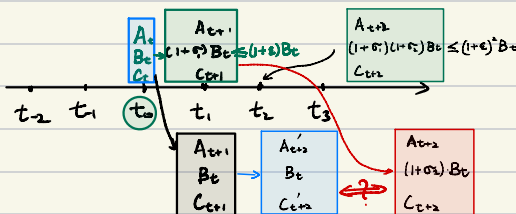
$$\frac{\partial A}{\partial C} \approx k_3 B \cdot \delta_t$$

TRY TO UNDERSTAND HOW 'E' IS GOING TO AFFECT OTHERS?

<fig. 1>

REAL

QSS of B



MASS
CONSERVATION
MAY GO WRONG

Trace Species A & C under the QSS assumption:

- Start from $[A_{t+1}, (1+\sigma) B_t, C_{t+1}]^T$
- <2> & <1> give the same updates for A & C
- Compare $[A_{t+2}, B_t, C_{t+2}]^T$ with $[A_{t+2}, (1+\sigma_2) B_t, C_{t+2}]^T$

Calculate $[A_{t+2}, (1+\sigma)B_t, C_{t+2}]^T$:

$$A_{t+2} = (1 - k_1 \delta_t) A_{t+1} + (1+\sigma) k_3 \delta_t B_t \cdot C_{t+1}$$

$$B_{t+2} = (1+\sigma) B_t$$

$$C_{t+2} \approx C_{t+1} + (1+2\sigma) k_2 B_t^2 \delta_t$$

Calculate $[A'_{t+2}, B_t, C'_{t+2}]^T$

$$A'_{t+2} = (1 - k_1 \delta_t) A_{t+1} + k_3 \delta_t B_t \cdot C_{t+1}$$

$$B'_{t+2} = B_t$$

$$C'_{t+2} = C_{t+1} + k_2 B_t^2 \delta_t$$

$$\begin{aligned} \begin{bmatrix} A_{t+2} - A'_{t+2} \\ B_{t+2} - B'_{t+2} \\ C_{t+2} - C'_{t+2} \end{bmatrix} &= \begin{bmatrix} \sigma k_3 \delta_t B_t C_{t+1} \\ \sigma B_t \\ 2\sigma k_2 B_t^2 \delta_t \end{bmatrix} \\ &= \begin{bmatrix} \sigma k_3 B_t \cdot C_{t+1} \\ B_{t+1} \\ 2\sigma k_2 B_t^2 \end{bmatrix} \cdot \delta_t \end{aligned}$$

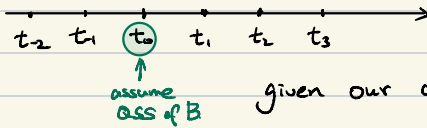
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Criteria 2 :

$$\begin{aligned} \frac{\sigma k_3 B_t \cdot C_{t+1}}{A_{t+1}} &\leq \varepsilon \\ \frac{2\sigma k_2 B_t^2}{C_{t+1}} &\leq \varepsilon \end{aligned}$$

TRY TO UNDERSTAND WHEN TO PAUSE THE QSSA

<fig.2>



<4>

given our criteria: $\delta_t \leq \epsilon \cdot \frac{B_t}{B_i}$

$$B \cdot \delta_t = \sigma B_t \quad (\sigma \leq \epsilon)$$

Real System:	QSSA System:
$B_{t+1} = (1 + \sigma) B_t$	$B_{t+1} = B_t$

$$B_{t+1, \text{QSS}} = \frac{1}{1 + \sigma} B_{t+1, \text{real}}$$

assume this σ is constant (or about the same)
for $T \in \{t_0, t_1, \dots, t_n, \dots\}$

for species B, given ' σ ', and the similarity pre-factor ' ϵ '
closer to one, more similar to the true system value

$$\left(\frac{1}{1 + \sigma}\right)^n = \epsilon$$

<5>

$$n = - \frac{\ln \epsilon}{\ln(1 + \sigma)}$$

of QSSA steps at the acceptable deviation level ' ϵ ' reached

To give a rough idea:

$$\epsilon = 0.9, \sigma = 1e-4 \rightarrow n \approx 1053 \text{ steps} \quad \epsilon = 0.99, \sigma = 1e-4 \rightarrow n \approx 100 \text{ steps}$$

$$\epsilon = 0.9, \sigma = 1e-3 \rightarrow n \approx 105 \text{ steps} \quad \epsilon = 0.99, \sigma = 1e-3 \rightarrow n \approx 10 \text{ steps}$$

???

CHALLENGE: After n steps, if I relax the QSS assumption, will the system converge to the 'true system' ???